

Benjamin Molnar
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SUMMARY

Highly motivated engineer interested in aerospace, robotics, and computer science

EDUCATION

Georgia Institute of Technology – 4.00 GPA | 2024-present **M.S. Computer Science – Machine Learning Specialization**

Currently completing master's degree in computer science

Focus on AI/ML techniques and their implementation in robotics applications

University of Maryland, College Park – 3.87 GPA | 2019-2021 **B.S. Mechanical Engineering, A. James Clark School of Engineering**

Dean's Scholarship, SME Scholarship

Completed undergraduate studies in 2.5 years

Designed tripteron – a 3D motion kinematics system for 3D printing applications

Designed autonomous large scale hydroponics system for urban use – Sustainability award winner

WORK EXPERIENCE

Advanced Manufacturing Engineer **Lightweight Innovations for Tomorrow (LIFT) | 2022-2024**

Led research to determine the best process parameters for novel alloys in the LPBF process

Designed network to virtually integrate machines, sensors, and microcontrollers with custom designed PCBs, including ESP32s, on manufacturing floor

Identified successful weld parameters by creating an algorithm to bulk extract bead characteristics from optical topography surface texture scans

Pathways Intern - Additive Manufacturing **National Institute of Standards and Technology (NIST) | 2019-2022**

Published original paper investigating thermography of overhanging metal AM parts in the NIST Journal of Research

Created executable app with GUI from MATLAB to process raw thermal camera data and output thermal videos, cooling rate plots, melt pool length graphs, etc.

Designed "Standard AM Kilogram" an impossible to traditionally manufacture part to showcase advantages of AM

Student Volunteer - Additive Manufacturing **National Institute of Standards and Technology (NIST) | 2018-2019**

Designed study to determine the effect of overhang angle on part quality during LPBF process

Coordinated use of high-speed thermal camera, visual light cameras, and thermocouples to collect in-situ process data for LPBF

EXTRACURRICULAR ACTIVITIES

SME – Life Member (2018 – Present), ASME – Member (2019 – Present), Volunteer Tutor (2013 – Present), FIRST (2018 – 2023), BSA – Eagle Scout (2011 – 2019), Cross Country (2013 – 2019)

SKILLS

NX, KiCAD, SOLIDWORKS, Inventor, COMSOL, SMD Soldering, Welding (MIG/TIG), C++, Python, MATLAB, PlatformIO, Arduino Framework, HTML/CSS/JS, Laser Powder Bed Fusion (LPBF), Fused Deposition Modeling (FDM), Stereolithography (SLA), Marlin, Photoshop, Premiere